Does Cycle Counting Reduce Inventory Error? A Review of the Evidence

Prepared by: Dr Imogen Ward, Logistics Research Group

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Abstract

Warehouse operators have long relied on annual physical inventories to correct stock records. Cycle counting, in which a small subset of items is counted on a rolling basis, is widely recommended as an alternative. This review summarises 14 studies published between 2009 and 2025 to assess whether cycle counting measurably improves inventory record accuracy, and under what conditions.

Background

Inventory record accuracy (IRA) is the proportion of stock records that match physical quantity within an agreed tolerance. Typical accuracy in manufacturing and distribution settings has been reported at 60 to 75 percent, which is far below the 95 percent or higher that most planning systems assume. Inaccurate records lead to stockouts, excess safety stock, and wasted picking effort.

Method

We searched four databases for studies reporting before and after accuracy for sites adopting cycle counting. Studies were included if they measured accuracy at item level and reported the counting frequency. Eleven studies were case studies and three were controlled trials. Quality was assessed using a simple three point scale.

Findings

Twelve of the 14 studies reported an improvement in IRA after cycle counting was introduced. The average improvement was 19 percentage points, with a range from 4 to 33. Sites that counted high-value items more often than low-value ones (an ABC approach) improved more than sites using a uniform schedule.

Three factors appeared to explain most of the variation. First, sites that investigated the cause of each discrepancy, rather than simply adjusting the record, improved faster. Second, blind counts, where the counter is not shown the expected quantity, reduced confirmation bias. Third, sites that triggered extra counts after exceptions, such as short picks, found errors sooner than sites that relied on the schedule alone.

Two studies found no significant benefit. Both were in sites with very high staff turnover where counters received little training.

Limitations

Most studies were observational and self-reported. Publication bias is likely, since operators with poor results may not publish them. Definitions of tolerance varied between studies, which makes direct comparison difficult.

Recommendations

Operators introducing cycle counting should:

- Classify items by value and count the highest-value group most often.
- Use blind counts by default.

- Investigate every discrepancy above tolerance and record the cause.
- Raise count tasks automatically after short picks and zero-stock confirmations.
- Train counters and keep turnover in the counting team low.

Further research

We recommend a controlled multi-site trial comparing uniform and ABC schedules, and further work on whether handheld prompts for root-cause entry improve the quality of investigation. A follow-up review should be completed by October 2027.